Inference rules : ways to get new expressions from known expressions.

$$\frac{\begin{matrix} P_1 \\ P_2 \\ \vdots \\ P_k \end{matrix}}{Q}$$

$P_1, P_2, \dots, P_k$ infer Q

if $P_1, P_2, \dots, P_k$ are T then Q is T

$P_1 \wedge P_2 \wedge \dots \wedge P_k \Rightarrow Q$ is a tautology

$P_1, P_2, \dots, P_k \vdash Q$

infer

Rule	tautology	name
$\dfrac{p}{p \vee q}$	$p \Rightarrow p \vee q$	addition
$\dfrac{p \wedge q}{p}$	$p \wedge q \Rightarrow p$	simplification
$\dfrac{\begin{matrix} p \\ p \Rightarrow q \end{matrix}}{q}$	$p \wedge (p \Rightarrow q) \Rightarrow q$	Modus Ponens
$\dfrac{\begin{matrix} \sim q \\ p \Rightarrow q \end{matrix}}{\sim p}$	$\sim q \wedge (p \Rightarrow q) \Rightarrow \sim p$	Modus Tollens
$\dfrac{\begin{matrix} p \Rightarrow q \\ q \Rightarrow r \end{matrix}}{p \Rightarrow r}$	$(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$	Hypothetical Syllogism
$\dfrac{\begin{matrix} p \\ q \end{matrix}}{p \wedge q}$	$p \wedge q \Rightarrow (p \wedge q)$	conjunction

example: with these hypothesis

H_1 - $\underbrace{\text{I woke up late}}_{w}$

H_2 - If $\underbrace{\text{I wake up late}}_{w}$ then $\underbrace{\text{I miss the bus}}_{b}$

H_3 - If $\underbrace{\text{I miss the bus}}_{b}$ then $\underbrace{\text{I walk to school}}_{s}$

Prove that $\underbrace{\text{I walked to school}}_{C \quad s}$

Proof:

1.	w	Hyp.
2.	$w \Rightarrow b$	Hyp.
3.	$b \Rightarrow s$	Hyp.
4.	b	M.P 1 + 2
5.	s	M.P 3 + 4

M. P.

$$\frac{\begin{array}{c} A \\ A \Rightarrow B \end{array}}{B}$$

exmp: with these hypothesis

- it is not $\underbrace{\text{sunny today}}_{p}$ and it is $\underbrace{\text{colder than yesterday}}_{q}$
- we will $\underbrace{\text{go swimming}}_{r}$ only if it $\underbrace{\text{is sunny}}_{p}$
- if we do not $\underbrace{\text{go swimming}}_{r}$ then we will $\underbrace{\text{take a cannoe trip}}_{s}$ $\Rightarrow$
- if we $\underbrace{\text{take a cannoe trip}}_{s}$ then we will $\underbrace{\text{be home by sunset}}_{t}$ $\Rightarrow$

prove that we will $\underbrace{\text{be home by the sunset}}_{t}$

Proof:

1. $\sim p \wedge q$ — Hyp.
2. $r \Rightarrow p$ — Hyp — $r \vee p \,/\, r \Rightarrow p$
3. $\sim r \Rightarrow s$ — Hyp.
4. $s \Rightarrow t$ — Hyp.
5. $\sim p$ — Simp (1)
6. $\sim p \Rightarrow \sim r$ — Contraposition (2)
7. $\sim r$ — M.P. (5) + (6)
8. s — M.P (7) + (3)
9. t — M.P (4) + (8)

6-7. $\sim r$ — M.T (5) + (2)

$$\frac{A \wedge B}{A} \quad \text{Simp}$$

$$\frac{\begin{array}{c} A \\ A \Rightarrow B \end{array}}{B} \quad \text{M.P.}$$

$$\frac{\begin{array}{c} \sim B \\ A \Rightarrow B \end{array}}{\sim A} \quad \text{M.T.}$$

if it is not sunny then we will not go swimming $\quad \sim p \Rightarrow \sim r$

$\equiv$

we will go swimming only if it is sunny $\quad r \Rightarrow p$

if it is sunny then we will go swimming

contraposition

Constructive Dilemma

$$\begin{array}{l} p \Rightarrow q \\ r \Rightarrow s \\ r \vee p \\ \hline s \vee q \end{array}$$

Proof:

✓1. $p \Rightarrow q$	Hyp	Imp $A \Rightarrow B \equiv \sim A \vee B$
✓2. $r \Rightarrow s$	Hyp	D.N. $A \equiv \sim\sim A$
✓3. $r \vee p$	Hyp	
✓4. $\sim\sim r \vee p$	Double Neg. (3)	Cont: $A \Rightarrow B \equiv \sim B \Rightarrow \sim A$
5. $\sim r \vee s$	Imp (2)	
✓6. $\sim r \Rightarrow p$	Imp (4)	
✓7. $\sim r \Rightarrow q$	H.S. (6) + (1)	
✓8. $\sim s \Rightarrow \sim r$	Contrapositive (2)	
✓9. $\sim s \Rightarrow q$	H.S. (7) + (8)	
✓10. $\sim\sim s \vee q$	Imp (9)	
✓11. $s \vee q$	Double neg. (10)	

We never used 5. So we can remove

Q. $\dfrac{p \Rightarrow q}{p \vee s \Rightarrow q \vee s}$ use constructive dilemma